The Limits of Predictions for Online Bipartite Matching: A Unified Experimental Study and a Budget–Stakes Law

Anonymous author
Anonymous affiliation

Abstract

Learning-augmented ("with-predictions") algorithms for online bipartite matching have proliferated: MinPredictedDegree consumes a per-node degree predictor, and a family of *test-and-fallback* schemes tests a type-histogram prediction on a sublinear prefix of the arrivals before committing to follow it or to fall back on an advice-free baseline. These algorithms have been analyzed in isolation — on different input models, against different error models, and largely in theory. We give the first unified experimental study, placing all of them on a single harness with a common structured prediction-error model, a common optimum, and confidence intervals, across synthetic graphs, six real-world graphs, and real request traces. Our central finding is that, on average-case (known-i.i.d.) inputs, the value of predictions is **robustness insurance, not a performance lever**: the advice-free baseline is already near-optimal, so the *consistency* upside of good advice is small, whereas unguarded prediction-following can crash far below the baseline — and the practical worth of the sophisticated algorithms is precisely that they never do. We then prove what the wall *costs*. For the test-and-fallback class we establish a sharp, two-sided **budget–stakes law**: deciding whether to follow the advice requires — and, via an explicit one-line rule, only requires — a prefix of length $\tilde{\Theta}(\theta/\delta^2)$, the inverse square of the stakes δ (what good advice gains over the baseline) scaled by the contention θ, independent of the instance length. The lower half binds *every* decision rule; the upper half refutes the natural conjecture that the hardness of tolerant distribution testing blocks all sublinear rules — the decision-relevant statistic is the *payoff* of following, which is exponentially cheaper to test than the prediction's distance. Because the stakes are capped by the baseline slack, on strong-baseline instances the price exceeds the horizon: upsides below $\approx \sqrt{(1-\rho_{\text{base}})/n}$ are uncapturable by any rule at any prefix length, and the upsides we measure sit in that range. Experiments and theory deliver one message: **on average-case matching, the advice's upside is smaller than the price of finding out whether to trust it.**

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1 Introduction

Online bipartite matching is the canonical model of irrevocable sequential allocation: offline resources are known in advance, requests arrive one at a time, and each request must be matched to an available neighbor — or dropped — immediately and irrevocably, before the rest of the input is seen. It underlies ad allocation, ride-hailing dispatch, organ exchange, and request routing in serving systems. Classical online algorithms — Greedy, KVV Ranking [11], and the stochastic-matching algorithms of Feldman et al. [7] and Jaillet–Lu [9] — are analyzed through the worst-case competitive ratio.

A now-large body of work augments these algorithms with a *prediction* about the input and asks for two guarantees at once: **consistency** (near-optimal performance when the prediction is good) and **robustness** (no worse than an advice-free baseline when the

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prediction is arbitrarily bad) [12, 14]. For online matching specifically, two strands have emerged. MinPredictedDegree (MPD) [1] matches each arrival to its available neighbor of minimum *predicted* degree, protecting scarce resources; it is robust by construction (a useless predictor reduces it to Ranking). A second strand — the *test-and-fallback* algorithms of Choo et al. [6] and Burathep–Erlebach–Moses [3] — is explicitly adaptive: it Mimics a type-histogram prediction on a sublinear prefix of arrivals, tests whether the observed arrivals match the prediction, and then either continues to follow it or falls back to Ranking.

These algorithms have been studied *in isolation* — each on its own input family, against its own error model, and largely through worst-case theory. As a result, three basic questions have no empirical answer. How do the algorithms actually compare, head to head, under one prediction-error model? How much does a prediction help, and at what cost, on realistic inputs? And — most importantly — *why* does the practical experience with these algorithms feel so different from their worst-case promise? This paper answers all three, with a unified experimental study and a theorem that explains what the study finds.

A unified experimental study. We place the advice-free baselines, the MPD family (and its Feldman/Jaillet–Lu augmentations), and the test-and-fallback algorithms on a single harness: common graph families, a common structured prediction-error model, a common optimum computed by Hopcroft–Karp, and 95% confidence intervals throughout. We also port the blind-follow-with-switching *combiner* of Chłędowski et al. [5] — the "cheap worst-case insurance" from the caching literature — and benchmark it (we do not claim it as new). The study runs across synthetic graphs spanning the difficulty range, six real-world graphs from the Network Repository, and real request traces.

The empirical wall. Across every setting we observe the same phenomenon, which we state as the paper's organizing thesis: *on average-case inputs the value of predictions is robustness insurance, not a performance lever.* Concretely (Section 3): the advice-free Ranking is already within a percent or two of the optimum, so the consistency upside of even perfect advice is small; unguarded prediction-following (blind Mimic, or MPD under adversarial predictions) crashes *below* the advice-free floor; and the entire practical value of the sophisticated algorithms is that they refuse to crash. Two distinct mechanisms deliver this — the structural robustness of the MPD-augmentations and the adaptive robustness of test-and-fallback — with different consistency/robustness trade-offs. We confirm the wall on the six real graphs and on real traces, where a cheap, non-ML historical predictor captures a meaningful fraction of the (small) oracle gap while never dropping below the baseline (Section 6). We also report a negative result: because MPD depends on its predictor only through the induced *order*, one might train the predictor with an order-aware loss instead of regression, but on realistic features the two coincide — reinforcing that the predictor, once order-faithful, is not the bottleneck.

Why the wall is necessary — and what it costs. The empirical wall is not an artifact of a particular generator or algorithm; it has a price tag. Our main theoretical contribution (Section 7) is a two-sided **budget–stakes law** for the test-and-fallback class. Informally:

> Deciding whether to follow the advice costs a prefix of length $k^* = \tilde{\Theta}(\theta/\delta^2)$ — the inverse square of the stakes δ (what good advice gains over the baseline), scaled by the contention θ. Below k^*, *no* decision rule is simultaneously consistent and robust; at k^*, an explicit one-line rule is.

The lower half is information-theoretic — it binds *any* measurable rule on the prefix, not merely the empirical-distance threshold used in practice — via a master consistency/robustness inequality and a Hellinger computation. The upper half is a **directional test**: classify each prefix arrival as agreeing or disagreeing with the advice's local prediction, and follow iff agreements win. Its analysis rests on a *payoff identity* — on the hard family, the advantage of following is exactly half the mean of a per-sample observable — which also refutes the tempting stronger conjecture (and an earlier version of our own claim) that tolerant-testing hardness [4] makes the decision impossible for every sublinear rule: the advice's *distance* to the truth is indeed hard to test, and the field's empirical-ℓ_1 thresholds are provably blind at sublinear prefixes, but the decision-relevant *payoff* is not. The wall then re-emerges exactly where the experiments live: stakes are capped by the baseline slack ($\delta \leq 2\varepsilon(1-\rho_{\text{base}})$), so on strong-baseline instances the budget k^* exceeds the horizon itself — every upside below $\Theta(\sqrt{(1-\rho_{\text{base}})/n})$ is uncapturable at *any* feasible prefix, and the upsides measured in Sections 3–6 sit in that range.

Contributions. - (C1) The first unified benchmark of learning-augmented online-matching algorithms — advice-free baselines, the MPD family, and the test-and-fallback schemes — on one harness with a common error model, a common optimum, and confidence intervals (Section 3). **- (C2) The robustness-insurance characterization.** Unguarded followers crash below the advice-free floor; two mechanisms (structural and adaptive) restore safety with distinct trade-offs; the consistency upside is small on average-case inputs, so the value is downside protection. Validated on synthetic graphs, six real graphs, and real traces (Sections 3, 6). **- (C3) An empirical engagement with the order-error theory.** MPD's loss is governed by a Kendall-τ order error onto which several error models collapse; we characterize the tightness and saturation of the known n−LIS bound [ACI22, App. D] rather than re-deriving order-dependence (Section 4). **- (C4) The first empirical study of test-and-fallback**, including a threshold-calibration pathology (a more accurate test can make a worse decision), its recalibration and resolution limit, and a benchmark of the dynamic combiner exhibiting an irrevocability penalty that explains why matching needs *test-then-commit* (Section 5). **- (C5) The budget–stakes law.** A sharp two-sided law for test-and-fallback: the follow/fallback decision costs a prefix of $\tilde{\Theta}(\theta/\delta^2)$ for *any* rule (lower bound), and an explicit directional test achieves it (upper bound). A payoff identity separates cheap payoff-testing from provably hard distance-testing — refuting, and reporting honestly, the natural tolerant-testing impossibility — and the corollary quantifies the wall: on strong-baseline instances, upsides below $\Theta(\sqrt{(1-\rho_{\text{base}})/n})$ are uncapturable at any prefix length (Section 7).

The experiments and the law are two views of one fact. The experiments discover a wall — predictions buy insurance, not performance; the law prices the wall — verifying advice costs the inverse square of its stakes, and average-case stakes cannot cover the bill. We adopt an AI-inference serving instantiation as a running application case study (Section 8) rather than a novelty claim, and we are careful throughout to credit the prior results our findings build on and to state the scope of each claim.

2 Setup

2.1 Model

We work in the **known-i.i.d.** model of online bipartite matching. There is a set R of n **offline resources** and a **type graph** on r **online types**, where type ℓ has a neighborhood $N(\ell) \subseteq R$ (the resources able to serve it). An **instance** is generated by drawing m arrivals i.i.d. from a distribution p over the r types; the i-th arrival reveals its type (hence its neighborhood) and must be matched, immediately and irrevocably, to a currently unmatched resource in its neighborhood, or left unmatched. The type graph and p are known to the algorithm; the realized arrival sequence is not. The classical setting is $r = n$ and $m = n$ (each offline vertex a distinct type); we also use *few-types* instances ($r \ll n$), where a prefix distribution-test is statistically meaningful.

We measure the **competitive ratio** $\rho(\mathrm{ALG}) = \mathbb{E}[|\mathrm{ALG}|]/\mathbb{E}[|\mathrm{OPT}|]$, where OPT is a maximum matching of the *realized* instance, computed exactly by Hopcroft–Karp [8]. All reported ratios are Monte-Carlo estimates with 95% confidence intervals (Section 2.5). The advice-free baseline we compare against throughout is KVV Ranking [11], whose ratio we denote ρ_{base} and call the **baseline strength**.

2.2 Algorithms

All the greedy-type algorithms share one primitive. Given a rank (a total order) on the resources, **GreedyWithPermutation** matches each arrival to its available neighbor of *minimum rank*. The advice-free baselines are three instances of this primitive together with the two stochastic-matching algorithms:

- **SimpleGreedy** — GreedyWithPermutation under the identity rank.
- **Ranking** [11] — GreedyWithPermutation under a uniformly random rank; the baseline.
- **Feldman** [7] and **Jaillet–Lu** [9] — the known-i.i.d. stochastic-matching algorithms (a max-flow blue/red decomposition, and a cap-3 LP realized by max-flow with per-type list distributions, respectively).

Degree predictions (object A). MinPredictedDegree (MPD) [1] is GreedyWithPermutation ranked by ascending *predicted degree* $\mu \in \mathbb{R}^R$: it matches to the available neighbor predicted to be rarest. With μ constant it reduces to Ranking; with μ the true realized degrees it is the consistency ceiling (MinDegree). Following ACI, we also form the **augmentations** Feldman(MPD) and JailletLu(MPD), which use the MPD rank only as the tie-break of the greedy fallback, so the worst-case-optimal base matching carries the load.

Type-histogram advice (object B) and test-and-fallback. The advice is a count vector $\hat{c} \in \mathbb{Z}^r_{\geq 0}$ over types (equivalently a distribution $q = \hat{c}/\|\hat{c}\|_1$). From $\hat{c}$ we build an *advice graph* ($\hat{c}_\ell$ copies of type ℓ), compute a maximum matching $\hat{M}$ on it, and record, per type, the resources $\hat{M}$ uses. **FollowPrediction** (blind Mimic) routes every arrival according to $\hat{M}$. **TestAndMatch** [6, 3] instead Mimics a sublinear prefix of k arrivals, estimates the ℓ_1 distance between the prefix type distribution and q, and — if the estimate is at most a threshold τ — continues to Mimic, else falls back to Ranking. The two variants differ only in an early-exit rule and in τ (Choo: $\tau = 2(\hat{n}/n - \beta)$; BEM: $\tau = 2(\hat{n}/n)(1-\beta)/(1+\beta)$), where β is the worst-case baseline ratio. Faithfulness note: the papers' ℓ_1 tester [10] has no deployable implementation and the authors themselves fall back to an empirical-ℓ_1 proof-of-concept; we do the same, and mark it as such.

The combiner (a benchmarked baseline, not a contribution). We port the blind-follow-with-switching combiner of Chłędowski et al. [5] — run FollowPrediction and Ranking as shadow experts and follow the leader with hysteresis — to contextualize test-and-fallback. We benchmark it; we do not claim it.

2.3 Prediction objects and the structured error model

To compare algorithms that consume *different* prediction objects (degree vectors μ vs type histograms $\hat{c}$), we corrupt each object with a per-family knob and report the families in parallel; this heterogeneity is intrinsic and is precisely why no unified benchmark existed.

For **degree predictions** we perturb the true degrees $\mu^\star$ with four structured error models, each returning both an ℓ_1 *magnitude* and a normalized **Kendall-τ order error** in $[0,1]$: *random-flip* (replace a fraction of entries by uniform noise), *systematic-bias* (a monotone rescale — order-preserving, hence Kendall-$\tau \equiv 0$ by construction), *adversarial* (reflect a fraction of entries, the most order-damaging), and *distribution-drift* (blend toward degrees from an independently drawn graph of the same family). Because MPD depends on μ *only through the induced order*, the order/magnitude distinction is not cosmetic — it is the axis Section 4 is about.

For **type-histogram advice** we perturb the realized counts $c^\star$ toward a concentrated (spiky) random target by a fraction $\eta \in [0,1]$ ($\eta = 0$: perfect advice, $\hat{c} = c^\star$; $\eta = 1$: advice unrelated to the truth), reporting the induced $\ell_1(p,q)$ as the advice-error axis for the test-and-fallback experiments.

2.4 Consistency and robustness

For a prediction-consuming algorithm we call ρ under *perfect* advice its **consistency** and its worst-case ρ over *adversarial* advice its **robustness**. A prediction algorithm is useful only if it is consistent *and* never falls (much) below ρ_{base}; the tension between the two, and its fundamental limits for the test-and-fallback class, are the subject of Sections 5 and 7.

2.5 Experimental methodology and reproducibility

The harness is a small dependency-light Python codebase (NumPy/SciPy/NetworkX). We draw all randomness from independent, reproducible streams via NumPy's spawn-based generators, and use **paired trials**: within a comparison, every algorithm and every error level reuses the same graph, arrival sequence, OPT, and tie-break seed, so differences are attributable to the prediction alone. Confidence intervals are 95% normal-approximation half-widths over trials. The Phase-2 portion reproduces a subset of Borodin, Karavasilis & Pankratov [2]; every figure and table in this paper is regenerated from a fixed seed by a single script, listed with its output in Appendix [REPRO].

3 The Unified Benchmark

This section places all three algorithm families on one harness and establishes the paper's organizing thesis. We report competitive ratios (mean $\pm$ 95% CI) as a function of prediction quality, in three panels chosen to span the regimes each family was designed for.

3.1 Design

The families consume different prediction objects, so we drive each with its own corruption knob and present them in parallel panels (Section 2.3); the shared harness — graphs, OPT, CI methodology, and the advice-free floor — is what makes the panels comparable.

- **Panel A — clvb_zipf** ($n = 1000$, 60 trials): heavy-tailed offline degrees, the regime where degree predictions carry signal.
- **Panel B — left-regular** d=5 ($n = 1000$, 60 trials): near-homogeneous degrees, where predictions have little to add.
- **Panel C — few-types** r=8 ($n = 2000$, 50 trials): near-perfect-matchable, the calibrated regime for the type-histogram test-and-fallback algorithms.

For the degree-prediction panels the quality columns are *perfect* (true degrees), *noisy* (random-flip at strength $\frac{1}{2}$), *adversarial* (order-reversing), and *garbage* (fully random ≡ Ranking predictor). For the advice panel the columns are $\eta \in \{0, 0.3, 0.6, 1.0\}$ (*perfect / mild / bad / garbage*). Confidence intervals are tight throughout (±0.001–0.003); we omit them from the prose and report them in Table 1.

Table 1 (the three panels, 95% CI) and **Figure 1** (grouped bars) are the section's data. The salient rows:

	perfect	noisy	adversarial	garbage
Panel A — clvb_zipf				
Ranking (floor)	0.948	—	—	—
MinDegree (oracle)	0.996	—	—	—
MPD	0.989	0.956	**0.908**	0.946
Feldman(MPD)	0.981	0.979	0.976	0.978
JailletLu(MPD)	0.977	0.976	0.974	0.975
Feldman / JailletLu (base)	0.887 / 0.901	—	—	—
Panel B — left-regular d=5				
Greedy = Ranking (floor)	0.890	—	—	—
MinDegree (oracle)	0.966	—	—	—
MPD	0.932	0.906	**0.854**	0.888
Feldman(MPD) / JailletLu(MPD)	0.906 / 0.903	0.903 / 0.902	0.896 / 0.898	0.900 / 0.901
Feldman / JailletLu (base)	0.758 / 0.789	—	—	—

	perfect	mild	bad	garbage
Panel C — few-types r=8				
Ranking (floor)	0.990	—	—	—
MPD (true degrees)	0.999	—	—	—
FollowPrediction	1.000	0.832	0.679	**0.472**
TestAndMatch (Choo)	1.000	0.984	0.989	0.990
TestAndMatch (BEM)	0.998	0.988	0.988	0.968
Combiner *(benchmark)*	0.990	0.990	0.990	0.990

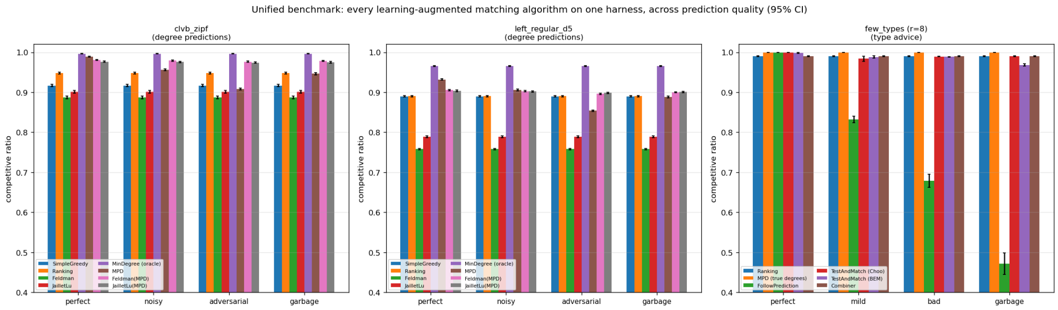

Figure 1 The unified benchmark (Table 1) as grouped bars: competitive ratio by algorithm and advice quality, one group per panel; the advice-free floor and oracle ceiling bracket every bar.

3.2 Four findings

(F1) Robustness is engineered, not free: naive followers crash below the floor. Both unguarded prediction-followers dive under the advice-free Ranking floor once the prediction is adversarial or garbage: MPD falls to $0.908 < 0.948$ (Panel A, adversarial) and $0.854 < 0.890$ (Panel B), and FollowPrediction collapses to $0.472 \ll 0.990$ (Panel C). A practitioner using either *unguarded* is strictly worse off than using no prediction at all. Every *robust* algorithm in the tables — the augmentations, TestAndMatch, the combiner — avoids this by construction, not by luck.

(F2) Two distinct robustness mechanisms, with different shapes. *Structural* robustness (Feldman(MPD), JailletLu(MPD)): a worst-case-optimal base matching carries the load and the prediction only breaks ties, so performance is nearly *flat* across quality — Feldman(MPD) moves only $0.981 \to 0.976$ from perfect to adversarial (Panel A). It cannot crash, but it also caps the upside (it never reaches the 0.996 oracle, and sits below MPD's 0.989 at perfect). *Adaptive* robustness (TestAndMatch): test a sublinear prefix, then commit — capturing the upside when advice is good (Choo 1.000 at perfect) *and* holding the floor when advice is bad (0.990 at garbage). On Panel C it is the only algorithm on the upper envelope at both ends. The two mechanisms trade consistency for robustness in opposite ways.

(F3) The consistency upside is small on average-case inputs; the spread lives on the bad-advice side. On few-types the advice-free Ranking is already 0.990 and MPD-with-true-degrees is 0.999 — under 0.01 for *any* advice to add on the good side. Every wide gap in Panel C ($1.000 \to 0.472$ for FollowPrediction; the half-wide envelope) is a *downside* gap. This is the thesis in one panel: learning-augmented matching on typical inputs is robustness insurance, not a performance lift — a fact Section 7 prices: safely capturing an upside this small would take a longer prefix than the instance itself.

(F4) The augmentation rescues structurally weak base algorithms. Feldman and Jaillet–Lu are tuned for the worst-case ratio and are the *weakest* advice-free entries on these average-case inputs (Panel B: 0.758 / 0.789, well below Greedy's 0.890). The MPD augmentation lifts them to ≈ 0.90 — the prediction does *more* for the worst-case-designed algorithms than for greedy. This pairing is visible only under a unified table.

3.3 Takeaway

Read together, F1–F4 say that on average-case matching the advice-free baseline is already near-optimal, unguarded prediction-following is unsafe, and the value of the sophisticated

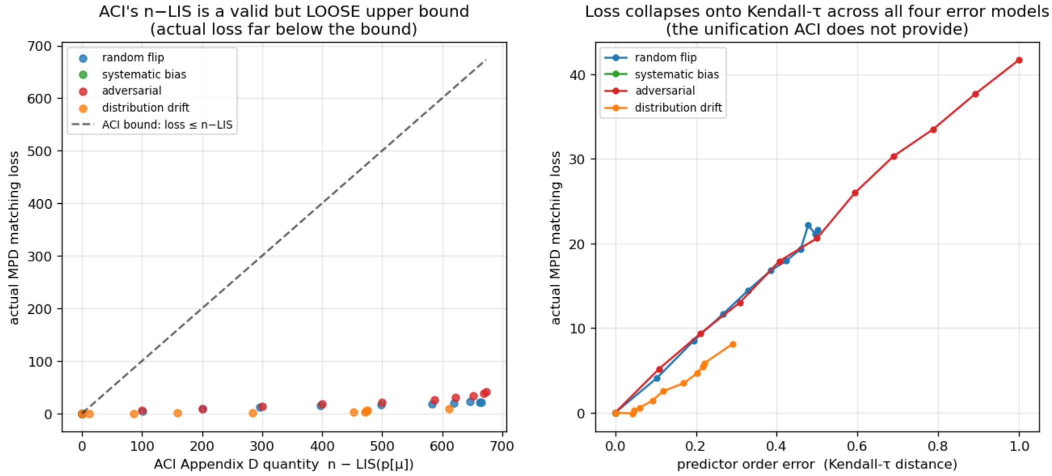

Figure 2 Order error, not magnitude, governs MPD's loss: (a) realized loss sits far below the $n - \mathrm{LIS}$ bound at every point; (b) the four error models collapse onto the Kendall-τ axis.

algorithms is downside protection delivered by one of two mechanisms. Sections 4–6 sharpen and stress-test each part — what governs the (small) loss, what the adaptive test costs, and whether the picture survives on real graphs, real traces, and a learned predictor — and Section 7 proves the wall is not an artifact but a price: a budget–stakes law that puts the required test prefix beyond the horizon exactly where the baseline is strong.

4 What Governs the Loss: Order-Error and ACI's Bound

MinPredictedDegree matches by ascending predicted degree, so it depends on the predictor μ *only through the order it induces on the resources.* Two predictors that induce the same order produce identical matchings; a monotone rescaling of μ changes nothing. The natural question is therefore not "how large is the prediction error" but "which *order* error governs the loss," and how tightly. This section answers that empirically. It also requires care: the qualitative fact that order — not magnitude — matters is *already theorem* in the literature, and we are explicit about what is prior and what is ours.

What is already known (ACI). Aamand, Chen & Indyk [ACI22, Appendix D] prove, for MinPredictedDegree on the CLV-B model, that the matching loss relative to the true expected degrees is at most $n-\mathrm{LIS}(p[\mu])$, where $p[\mu]$ is the true weights ordered by μ and LIS is the longest non-decreasing subsequence — a pure *order* quantity. In particular a monotone (order-preserving) predictor has $p[\mu]$ already sorted, so $\mathrm{LIS} = n$, $n - \mathrm{LIS} = 0$, and the loss is zero. Thus *order-dependence* and *the zero-effect of a monotone bias* are ACI's results, not ours; our `systematic_bias` error model (a monotone rescale) has Kendall-$\tau \equiv 0$ *by construction* and, consistently, leaves MPD's ratio exactly unchanged across the benchmark (Section 3) — an empirical confirmation of ACI's statement, not a new finding.

Our contribution is to characterize *how the bound behaves* and *which order measure predicts the realized loss.* We sweep the four structured error models across strength and record, per model and level, three quantities on the same instances: the actual MPD matching loss, ACI's $n - \mathrm{LIS}$, and the normalized Kendall-τ order error (**Figure 2**; $n = 1000$, Zipf exponent 1.0, 40 trials).

(i) ACI's $n - \mathrm{LIS}$ bound is correct but very loose. The realized loss sits far below the $n-\mathrm{LIS}$ upper bound at every point — by roughly 16× (adversarial) to 75× (distribution-drift). Figure 2(a) plots loss against $n-\mathrm{LIS}$ with the $y=x$ bound; all points lie close to the axis.

(ii) $n-\mathrm{LIS}$ **saturates and cannot distinguish error structures.** For every non-trivial error the quantity is pinned near its maximum ($n-\mathrm{LIS} \approx 610$–$673$ for $n=1000$), even though the true losses differ by a factor of five across models — 8.1 (drift), 21.6 (random-flip), and 41.7 (adversarial) at full strength. As an upper bound that is (almost) always $\approx n$, it carries little information about which prediction is actually more harmful.

(iii) Kendall-τ is the governing order measure, and the error models collapse onto it. Plotting the realized loss against the Kendall-τ order error (Figure 2(b)), the four structured error models fall on a single increasing curve: loss rises monotonically with Kendall-τ ($0.29 \to 0.50 \to 1.0$ for drift, random-flip, adversarial, tracking $8.1 \to 21.6 \to 41.7$ in loss), with `systematic_bias` pinned at $\tau=0$ and zero loss. The quantity that *predicts* the loss — and unifies the error models — is the Kendall-τ order distance, which $n-\mathrm{LIS}$ (saturated) cannot resolve.

Takeaway (and scope). We do not claim that order, rather than magnitude, governs MPD — ACI's Appendix D establishes that, and the zero-effect of a monotone bias with it. What we add is a precise empirical characterization: ACI's $n-\mathrm{LIS}$ bound is loose by one to nearly two orders of magnitude and saturates, whereas Kendall-τ predicts the realized loss and unifies the four error structures onto one curve. This both sharpens the theory's practical content and justifies the single order-error axis we use when we stress-test the predictor on real data (Section 6).

5 Test-and-Fallback in Depth

The test-and-fallback algorithms are the paper's adaptive robustness mechanism, and the object of the theory in Section 7. This section gives their first empirical study: the robustness envelope they achieve, a counter-intuitive failure of the acceptance threshold, its recalibration, and the resolution limit that recalibration exposes — the empirical face of the budget–stakes law we prove in Section 7. Throughout, the ℓ_1 test is the empirical-ℓ_1 proof-of-concept the original authors also fall back to (Section 2.2).

5.1 The robustness envelope

We sweep advice error $\ell_1(p,q)$ on few-types instances ($n=2000$, $r=8$, prefix $k=200$, 40 trials) and compare blind FollowPrediction, TestAndMatch (Choo and BEM variants), and the Ranking floor (**Figure 3**). The picture is textbook. FollowPrediction degrades *linearly*, from 1.000 at perfect advice to 0.453 at $\ell_1 \approx 1.1$ — well *below* the advice-free floor (≈ 0.99). TestAndMatch instead stays on the **upper envelope**: it captures the benefit when advice is good and, when advice is bad, its prefix test rejects it and it falls back to Ranking, never crashing (BEM $0.998 \to 0.969$; Choo $1.000 \to 0.991$ across the same sweep). This is the adaptive counterpart of the structural robustness of Section 3: the engineered mechanism that MPD lacked.

5.2 A threshold that is too lenient on average-case inputs

What does the "test then maybe fall back" machinery cost, and does a better test make a better decision? We sweep the prefix (testing) size k at *borderline* advice ($\eta=0.15$, true $\ell_1 \approx 0.16$) — the regime where the test must work hardest (**Figure 4**). Counter-intuitively, **a larger,**

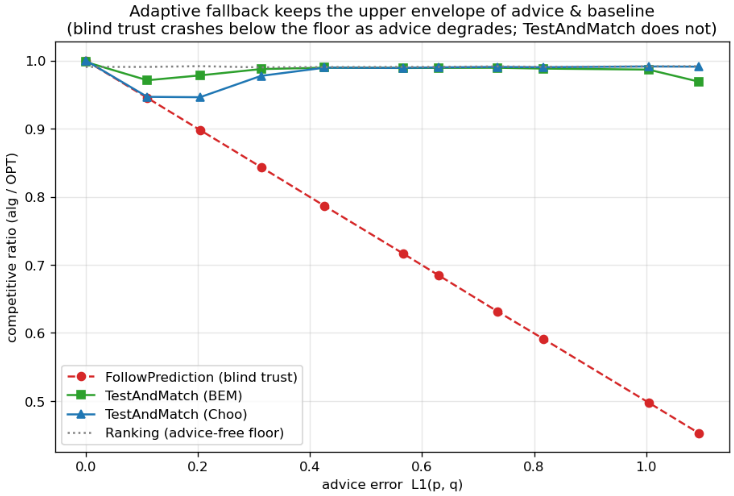

Figure 3 The robustness envelope: blind FollowPrediction crashes below the advice-free floor as advice error grows; TestAndMatch stays on the upper envelope.

more accurate test makes the *worse* decision: as k grows $25 \to 100 \to 200 \to 800$, the ratio *falls* $0.992 \to 0.981 \to 0.969 \to 0.956$ and the misjudgement rate (against an empirical oracle: should we have followed, given that following actually underperformed the baseline here?) *rises* $0.00 \to 0.13 \to 0.33 \to 0.60$.

The mechanism is a real and reportable finding. The Choo/BEM acceptance threshold τ is calibrated to the *worst-case* baseline $\beta \approx 0.696$. But on these average-case instances the baseline is ≈ 0.99 (Section 3, F3), so the *empirical* break-even — the advice error at which following stops beating the baseline — sits at $\ell_1 \approx 0$, far below τ. A small, noisy prefix over-estimates ℓ_1 and *accidentally rejects* the borderline advice, landing on the strong Ranking floor; a large, accurate prefix correctly measures $\ell_1 \approx 0.16 < \tau$ and so **accepts** the mildly-bad advice the worst-case threshold deems acceptable — and underperforms the baseline. On strong-baseline inputs the worst-case-calibrated threshold is too *lenient*, and improving the test's accuracy only makes the algorithm follow that miscalibrated threshold more faithfully.

5.3 Recalibration, and the resolution limit it exposes

The natural fix is to recalibrate τ to the *measured* baseline $\hat{\beta}$ (the mean empirical Ranking ratio on a small calibration sample) rather than the worst-case β. On the same instances this eliminates the pathology: at borderline advice the worst-case threshold's misjudgement climbs $0.03 \to 1.00$ as the prefix grows and its ratio drops to 0.920, whereas the recalibrated threshold holds misjudgement $= 0.00$ at every prefix size and ratio ≈ 0.986 — a larger prefix no longer hurts.

But recalibration exposes a deeper, honest limit. At *perfect* advice ($\ell_1 = 0$) the worst-case threshold scores 1.000 (it accepts and follows), while the recalibrated one scores only 0.987: the recalibrated threshold $\tau \approx 2(1-\hat{\beta}) \approx 0.028$ is *smaller than the empirical-ℓ_1 estimator's own noise floor* (≈ 0.05–0.13 at this prefix and support), so the test can never confidently *accept* — it rejects everything, including perfect advice, and effectively always plays the baseline. In other words:

On strong-baseline instances, no practical empirical-ℓ_1 threshold can both capture the

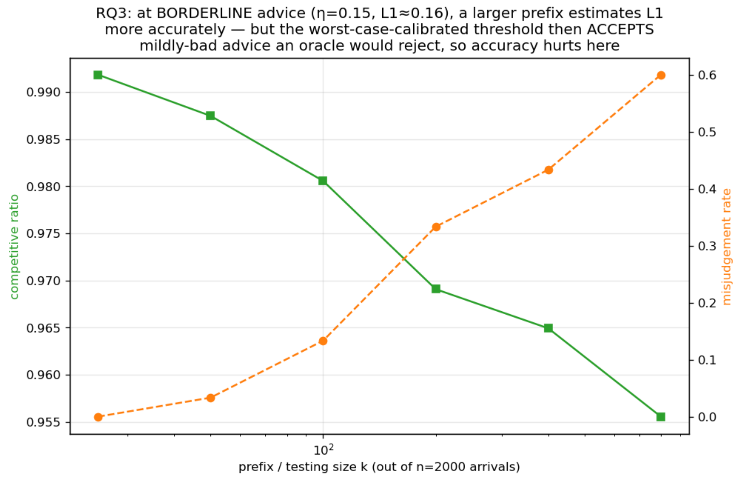

Figure 4 Testing cost at borderline advice: under the worst-case-calibrated threshold, a larger and more accurate prefix test makes the *worse* decision.

consistency upside and stay safe: the upside is tiny (the baseline is already $\approx$ OPT) and sits *below the estimator's resolution*. The worst-case threshold over-accepts (§5.2); the recalibrated threshold over-rejects; and a better tester only follows whichever threshold more faithfully.

Section 7 turns this phenomenon into a two-sided law. The resolution limit above is specific to the empirical-ℓ_1 statistic — §7.7 shows that class is blind for a structural reason (the plug-in distance saturates at $k \ll r$ regardless of advice quality) — but the deeper point holds for *any* test on a sublinear prefix: the follow/fallback decision costs a prefix of $\tilde{\Theta}(\theta/\delta^2)$, and on strong-baseline instances like these the stakes δ are so small that the price exceeds the horizon. Section 5 is the empirical shadow; Section 7 is the law.

5.4 The dynamic combiner is dominated, and shows why matching needs test-then-commit

Finally we benchmark the Chłędowski-style dynamic combiner (Section 2.2) to contextualize the *commit-once* structure of test-and-fallback. In its intended robust tuning the combiner sits exactly on the floor (0.990 across all advice quality on few-types): it never crashes, but captures *none* of the consistency upside — strictly dominated by TestAndMatch on the good-advice side and equal to it on the bad side.

More instructively, an *eager* combiner that switches experts mid-stream reveals a penalty specific to irrevocable problems. With perfect advice, eager switching scores 0.927 — *below both* the pure follower (1.000) and the pure baseline (0.958; verified in `tests/test_combiner_small.py`) — because switching from Ranking to Mimic mid-run lands the committed matching in an *incompatible hybrid*: the Mimic plan's slots are already half-consumed by the Ranking phase, and the Ranking progress is abandoned. In an irrevocable problem the follow/fallback decision must be made *before* the bulk of the commitments, not adapted across them — which is precisely why Choo/BEM test on a prefix and then *commit*, rather than switching dynamically as the caching-style combiner does. The dynamic combiner that is cheap worst-case insurance for caching does not port cleanly to matching, and the unified benchmark shows why.

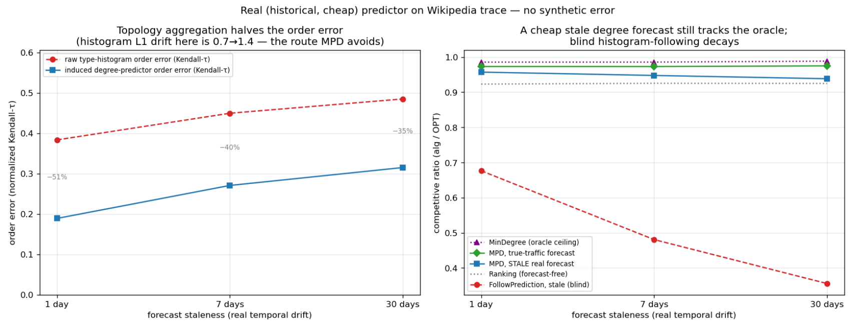

Figure 5 A cheap historical-count predictor on the Wikipedia trace: near-oracle ratio at every staleness level, never below the advice-free floor.

6 External Validity

Sections 3–5 are on synthetic graphs with a synthetic error knob. Is the picture real? We stress it three ways: a genuine, cheap predictor on a real request trace (§6.1); the six real-world graphs (§6.2); and whether *learning* the predictor changes anything (§6.3).

6.1 A real, cheap predictor

We replace the synthetic error knob with the cheapest realistic predictor: last-window historical statistics. Real Wikipedia daily pageviews give a live day (the truth) and earlier days (a 1-, 7-, or 30-day-stale forecast), so the prediction error is genuine temporal drift; we map the trace onto a fixed serving topology and consume the forecast through the degree route (MPD) (**Figure 5**; `docs/REAL_PREDICTOR.md`).

Three facts emerge, all favorable and all honest. First, **the cost premise does not bite**: the predictor is a linear-time count ($\mu = A\,p$ over historical frequencies), 0.108 ms per instance — about 2% of the cost of computing OPT once — not an ML inference. Second, **the benefit is real, partial, and never harmful**: a stale forecast captures 27%–68% of the achievable oracle gap (falling with staleness) and *always* stays above the forecast-free baseline (0.938–0.957 vs 0.923–0.925, even at 30 days). Third, and explaining why, **topology aggregation makes the cheap predictor order-faithful**: the induced degree predictor's order error is only Kendall-$\tau \approx 0.19$–0.32, roughly *half* the raw type-histogram's (0.38–0.49) — and since MPD depends only on order (Section 4), the aggregated degree route survives real drift. Consuming the *same* stale forecast through the raw histogram instead is catastrophic: blind FollowPrediction collapses to $0.68 \to 0.36$, far below the ≈ 0.92 baseline — exactly what the robust algorithms of Sections 3 and 5 are for.

6.2 Six real-world graphs

We re-run the degree-prediction roster of Section 3 on the six Network-Repository graphs (two Facebook social, two C. elegans biological, two economic input-output), across the same quality columns, with 95% CIs (**Figure 6**; `docs/REALWORLD_ROBUSTNESS.md`).

The two load-bearing findings are universal. **F1 holds on all six graphs**: naive MPD fed an adversarial predictor falls *below* the Ranking floor everywhere — by 0.07 (Caltech36: $0.843 < 0.913$) to 0.11 (CE-PG: $0.782 < 0.883$). **F3 is universal and confirms its own logic**: the consistency upside (best prediction algorithm at perfect advice, over the floor) is small everywhere (mean +0.049; range +0.022–+0.077), and it is *smallest exactly where the*

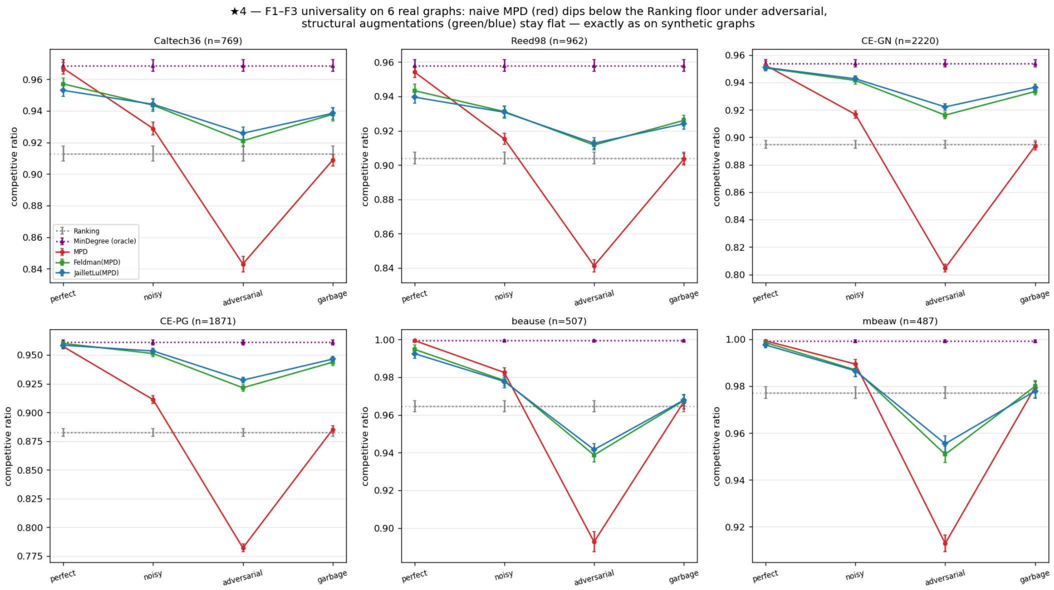

Figure 6 The six real-world graphs: unguarded following falls below the advice-free floor on every graph, and the augmentation restores safety — the benchmark's findings are universal, not generator artifacts.

baseline is strongest — the two dense economic graphs, with Ranking already 0.965/0.977, give the tiniest upsides (+0.035/+0.022).

The structural-robustness finding **F2 holds qualitatively on all six** (the augmentations are always less sensitive to prediction quality than naive MPD) and *strictly* on the four social/bio graphs (augmentation spread 0.22–0.29× MPD's); on the two dense economic graphs the protection is only *partial* — the augmentation cushions the adversarial drop (beause: 0.939 vs naive MPD's 0.893) but cannot fully clear the unusually high 0.965 floor, dipping ≈ 0.03 below it (spread ratio 0.53–0.55). This econ boundary is instructive rather than a failure: those graphs are so dense that matching is nearly trivial (Ranking ≈ 0.97, MinDegree = 1.00), so there is neither upside to capture (F3) nor much downside to protect — the mechanisms compress together where robustness does not matter. Finally, F4 is *dramatic* here: the worst-case-designed Feldman/Jaillet–Lu are the weakest advice-free entries on the econ graphs (0.73–0.77), and the MPD augmentation lifts them to 0.99 — a +0.26 rescue.

6.3 Does *learning* the predictor help? A negative result

Because MPD consumes the predictor only through its order (Section 4), one might train the predictor with an order-aware (rank) loss rather than regression. We test this and report an honest negative (`docs/RANK_LEARNING_M0_M3.md`). With *deliberately divergent* synthetic features — one carrying magnitude, one carrying order — rank-training does beat regression sharply (matching ratio 0.989 ≈ oracle vs 0.974, with rank-training having *worse* regression error but better order). But the advantage is gated by two things that both vanish in practice: it is zero when features do not induce an order/magnitude divergence and zero on easy instances (F3 again), peaking at only +1.3% on synthetic graphs; and, decisively, **on real temporal features it disappears** — trained on lagged counts from real serving traces, the rank- and regression-trained predictors produce *identical* order (Kendall-τ 0.126 vs 0.126) and identical matching ratio, across every topology and lag we tried. The dissociation that powers order-aware training is a property of engineered features, not of realistic ones. The lesson reinforces the thesis: once a predictor is order-faithful — which a cheap historical count already is (§6.1) — neither a better algorithm nor a better-trained predictor buys much on average-case matching.

6.4 Summary

On real predictors, real graphs, and a learned predictor, the same wall stands: the advice-free baseline is near-optimal, unguarded following is unsafe, and predictions buy downside protection rather than performance. Having established the wall empirically across every setting we could reach, we now prove it is necessary.

7 A Budget–Stakes Law: What a Sublinear Test Can and Cannot Buy

Every experiment above met the same wall: on average-case matching the advice-free baseline is near-optimal, so predictions buy robustness insurance rather than performance, and — for the test-and-fallback algorithms — no acceptance threshold both captures the (tiny) upside and stays safe (§5.3). This section quantifies the wall. The result is a two-sided **budget–stakes law**: on a natural instance family, deciding whether to follow the advice requires a prefix of length $k^* = \tilde{\Theta}(\theta/\delta^2)$ — the inverse square of the stakes δ, scaled by the contention θ — and this is *sharp*: below k^* **no** decision rule whatsoever can be simultaneously consistent and robust (Theorem 1(i)), while at k^* an explicit, embarrassingly simple rule succeeds (Theorem 1(ii)). The corollary is the wall: stakes are capped by the baseline slack, so on strong-baseline instances the required prefix exceeds the entire horizon — any upside smaller than $\approx 2\sqrt{(1-\rho_{\text{base}})/n}$ is uncapturable at *every* $k \leq n$, and the upsides we measured in §3–§5 sit in exactly that range.

En route we record an honest negative of independent interest (§7.7): the natural conjecture — that the $\tilde{\Theta}(r/\log r)$ hardness of *tolerant distribution testing* [4, 13, 10] makes the follow/fallback decision itself hard for any rule at sublinear k — is **false**. The advice's *distance* $\ell_1(p, q)$ is indeed hard to test, and the empirical-ℓ_1 rules the field actually uses [6, 3] are provably blind at $k \ll r$; but the decision-relevant functional is the *payoff* of following, which on decomposable instances collapses to a per-sample observable. Testing the payoff is exponentially cheaper than testing the prediction.

Relation to prior work. Choo et al. [6] and Burathep–Erlebach–Moses [3] give *upper* bounds (algorithms) in this model and no statistical lower bound; Choo et al.'s Thm 3.1 is the generic adversarial trade-off (no algorithm is 1-consistent and $> \frac{1}{2}$-robust), driven by adversarial instances, not by the resolution of a sublinear test. Their acceptance threshold $\tau = 2(\hat{n}/n - \beta) - \varepsilon$ couples to the baseline β *constructively*; Theorem 1(i) is the information-theoretic counterpart — it binds *every* measurable rule on the prefix. The consistency/robustness lower bounds elsewhere in the with-predictions literature [14] are problem-intrinsic Pareto frontiers, not gated by a test. Tolerant-testing sample complexity [4] enters our story twice, both times off the critical path: it explains *why* the field's ℓ_1-threshold rules are blind (§7.7), and its hard instances are exactly what our payoff identity (Lemma 2) sidesteps. To our knowledge, both the budget–stakes law and the observation that payoff-testing strictly separates from distance-testing in an online algorithm are new; a dedicated literature pass on payoff-estimating acceptance rules is in progress.

7.1 The test-and-fallback class

Fix an instance family and a budget $k = o(n)$. A **test-and-fallback** algorithm A_k observes the first k arrivals $X_{1:k}$ and the advice σ, outputs a (possibly randomized) decision $D \in \{\mathsf{Follow}, \mathsf{Fallback}\}$ as a function of $(X_{1:k}, \sigma)$, and thereafter Mimics σ if $D = \mathsf{Follow}$ and runs the advice-free baseline B (Ranking) otherwise. Because $k = o(n)$, the prefix's own contribution to the ratio is $O(k/n) = o(1)$, absorbed below. Write $\rho_{\text{base}} = \mathbb{E}[B]/\text{OPT}$ for the baseline strength, **consistency** for the ratio under good advice, and **robustness** for the ratio under adversarial advice.

7.2 The master trade-off (rigorous)

Lemma 1. Let G, Bd be two instance distributions sharing the *same* advice σ, with prefix laws $\mathcal{L}_G, \mathcal{L}_{\text{Bd}}$ and $\gamma_k := \text{TV}(\mathcal{L}_G, \mathcal{L}_{\text{Bd}})$. Suppose following σ gains δ under G and loses Δ under Bd relative to B. Parameterize A_k by the fraction η_c of the upside it forgoes under G and its robustness loss $\eta_r \Delta$ under Bd. Then

$$(1 - \eta_c) \;\le\; \eta_r + \gamma_k + o(1).$$

Proof. Conditioning on D under G, $\mathbb{E}_G[A_k]/\text{OPT} = P_G(\mathsf{F})\,\mathbb{E}_G[\text{Mimic}]/\text{OPT} + P_G(\mathsf{R})\rho_{\text{base}} \pm o(1) \ge \rho_{\text{base}} + \delta\, P_G(\mathsf{F}) - o(1)$, so the captured upside $(1-\eta_c)\delta$ forces $P_G(\mathsf{F}) \ge 1 - \eta_c - o(1)$. Symmetrically $P_{\text{Bd}}(\mathsf{F}) = \eta_r + o(1)$. Since D is a function of $(X_{1:k}, \sigma)$ and σ is identical, $|P_G(\mathsf{F}) - P_{\text{Bd}}(\mathsf{F})| \le \gamma_k$; chaining gives the claim. ◀

With an *uninformative* prefix ($\gamma_k = o(1)$) one cannot be both near-fully-consistent ($\eta_c \to 0$) and robust ($\eta_r \to 0$). Everything below is a computation of how large k must be before the prefix becomes informative — and (the half the earlier draft missed) of how *small* a k already suffices.

7.3 The construction and the affine law

The family is a disjoint union of m independent **rare-resource cells**, each over a distinct type-pair, so the support is $r = \Theta(m)$ and (at $m = \Theta(n)$) each type is seen $O(1)$ times. A cell has resources $\{a_i, b_i\}$, a flexible request (neighborhood $\{a_i, b_i\}$) that must be routed before a specialist ($\{a_i\}$ or $\{b_i\}$, present with probability θ) is seen; the right route depends on the future specialist. In closed form (verified numerically), per cell $\text{OPT} = 1 + \theta$, baseline $= 1 + \theta/2$, and following the advice gains or loses exactly $\theta|s - \frac{1}{2}|$ over the baseline according to whether the advice's *direction* is right, where $s = \frac{1}{2} \pm \varepsilon$ is the specialist bias and ε the signal; the advice-to-truth distance is $\ell_1 = 2\theta|s - \hat{s}|$ per cell.

These aggregate cleanly. Summing over cells yields an **exact affine law** (proven; verified to three decimals):

$$\mathbb{E}[\text{follow-ratio}] \;=\; \rho_{\text{perfect}} - \tfrac{1}{2}\,\ell_1(p, q), \qquad \ell_1^\star \;=\; 2\big(\rho_{\text{perfect}} - \rho_{\text{base}}\big),$$

where ρ_{perfect} is the all-correct-direction ratio and $\ell_1^\star$ the break-even. Two identities anchor what follows: the baseline slack is $1 - \rho_{\text{base}} = \frac{\theta}{2(1+\theta)}$, and the maximum stakes (the full upside) are

$$\rho_{\text{perfect}} - \rho_{\text{base}} \;=\; \frac{\theta\varepsilon}{1+\theta} \;=\; 2\varepsilon\,(1 - \rho_{\text{base}}):$$

the upside is an ε-fraction of the baseline slack. The canonical scenario pair takes a wrong-direction fraction $\varphi = \frac{1}{4}$ (good: $\ell_1 = a$, gain δ) versus $\varphi = \frac{3}{4}$ (bad: $\ell_1 = b$, loss Δ), with $\delta = \Delta = \ell_1^\star/4$.

7.4 The payoff identity: the stakes are a per-sample observable

The earlier route tried to make the follow/fallback decision *hard* by making $\ell_1(p,q)$ hard to test. On this family that cannot work, for a reason worth a lemma. Classify each arrival by its role in its own cell, relative to the advice's predicted direction:

$$c(X) \;=\; \begin{cases} +1 & X \text{ is the advice-favored specialist of its cell,} \\ -1 & X \text{ is the disfavored specialist,} \\ 0 & X \text{ is flexible.} \end{cases}$$

Lemma 2 (payoff identity). For every type distribution p in the family,

$$\mathbb{E}_p[c] \;=\; 2\cdot\frac{\mathbb{E}_p[\mathrm{Mimic}]-\mathbb{E}_p[B]}{\mathrm{OPT}}.$$

The expected advantage of following the advice is half the mean of a bounded, per-sample observable. (Verified to three decimals across all wrong-fractions; `scripts/verify_witness_gap.py`.)

Proof. The favored and disfavored specialists of cell i have masses $\theta(\frac{1}{2}\pm\varepsilon d_i\hat{d}_i)/N$, where $d_i,\hat{d}_i$ are the truth and advice directions and $N=m(1+\theta)=\mathrm{OPT}$; so cell i contributes $2\theta\varepsilon d_i\hat{d}_i/N$ to $\mathbb{E}[c]$, while by the cell constants its follow-advantage is $\theta\varepsilon d_i\hat{d}_i$. Sum over cells. ◀

7.5 The budget–stakes law

Theorem 1 (budget–stakes law). On the cell family with contention θ, signal ε, and the canonical scenario pair with stakes $\delta=\Delta$:

(i) (Impossibility below the budget — any rule.) If $k=o\big(\theta/\delta^2\big)$ *(equivalently* $o\big((1+\theta)/(\theta\varepsilon^2)\big)$*)*, then every test-and-fallback algorithm A_k, deciding by *any* measurable rule on the prefix, has $\eta_c+\eta_r\geq 1-o(1)$: it forgoes the upside or eats the loss.

(ii) (The budget suffices — an explicit rule.) The **directional test** — follow iff $\sum_{j\leq k}c(X_j)>0$ — achieves $\eta_c+\eta_r\leq\epsilon_0$ once $k\geq C\,(\theta/\delta^2)\log(1/\epsilon_0)$. In particular k is *independent of* n for constant error.

Hence the critical budget is $k^*=\tilde{\Theta}(\theta/\delta^2)$.

Proof of (ii). Under the good scenario $\mathbb{E}[c]=2\delta$ (Lemma 2), under the bad -2Δ; $\mathrm{Var}(c)\leq\mathbb{P}(\text{specialist})=\theta/(1+\theta)$; Bernstein's inequality gives error $\exp(-\Omega(k\delta^2/\theta))$ on both sides. ◀

Proof sketch of (i). Couple the two scenarios so their direction vectors differ on a $\frac{1}{2}$-fraction of cells. For a coupled pair of fixed direction vectors, the per-sample laws differ only on the specialists of differing cells, giving squared Hellinger distance $H^2_{\mathrm{per}}=\Theta(\theta\varepsilon^2)$; by tensorization and joint convexity, $\gamma_k\leq\sqrt{2k\,H^2_{\mathrm{per}}}=o(1)$ whenever $k=o(1/(\theta\varepsilon^2))$. Lemma 1 then forces $\eta_c+\eta_r\geq 1-o(1)$. Since $\delta=\Theta(\theta\varepsilon)$, the two thresholds match up to the stated factors. ◀

Numerically (§`verify_witness_gap.py`, $\theta=0.6$, $\varepsilon=0.3$, $\delta=0.056$): the directional test reaches $\eta_c+\eta_r\approx 0.06$ at $k=100$ and 0.007 at $k=200$, and stays flat as n grows from 3,200 to 320,000 at fixed $k=200$ — the budget really is n-free once the stakes are constant.

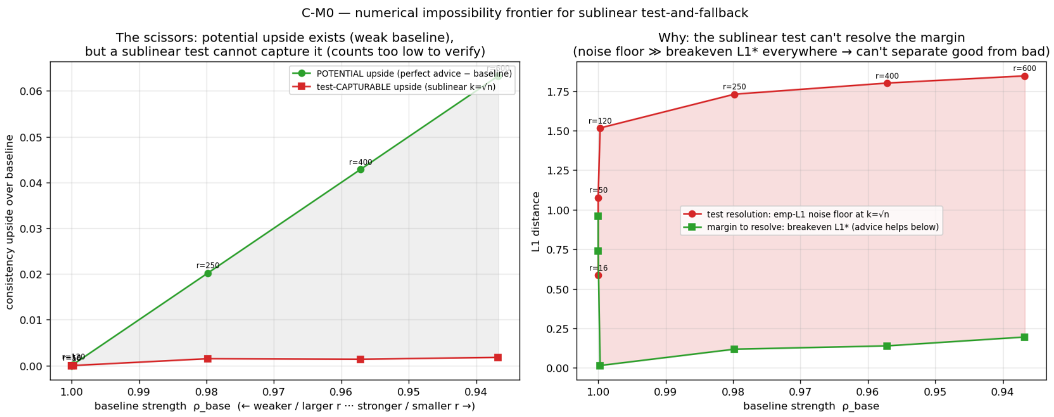

Figure 7 The budget–stakes scissors: as the baseline weakens (left), the potential upside of perfect advice grows, while the upside a feasible test can safely capture stays pinned near zero — the empirical test's resolution sits far above the break-even margin wherever the upside exists.

7.6 The wall, quantified: strong baselines price the test out of the horizon

Stakes are capped by the slack: $\delta \le \rho_{\text{perfect}} - \rho_{\text{base}} = 2\varepsilon(1-\rho_{\text{base}})$. Substituting into Theorem 1(i), capturing *any* constant fraction of the maximum upside requires

$$k \;\ge\; \tilde{\Omega}\Big(\frac{\theta}{\varepsilon^2(1-\rho_{\text{base}})^2}\Big) \;=\; \tilde{\Omega}\Big(\frac{1}{\varepsilon^2\,(1-\rho_{\text{base}})}\Big),$$

using $\theta = \Theta(1-\rho_{\text{base}})$. The wall appears when this exceeds the horizon itself: $k^* \ge n$ exactly when the upside $2\varepsilon(1-\rho_{\text{base}})$ falls below $\approx 2\sqrt{(1-\rho_{\text{base}})/n}$. In words:

Corollary (uncapturable upsides). On strong-baseline families, every advice upside smaller than $\Theta\big(\sqrt{(1-\rho_{\text{base}})/n}\big)$ is uncapturable by any test-and-fallback rule at any prefix length $k \le n$ — the information needed to decide does not exist inside the instance.

At the parameters of our benchmark (§3: $\rho_{\text{base}} \approx 0.99$, $n = 2000$) the threshold is ≈ 0.004, and the measured upsides are < 0.01 (F3) — the empirical wall sits in the regime the corollary governs. This is the **scissors** of **Figure 7** (`results/impossibility_frontier.png`), now with the correct mechanism: the *potential* upside (perfect advice minus baseline) grows as the baseline weakens, while the upside a feasible test can safely capture is pinned by the $\sqrt{(1-\rho_{\text{base}})/n}$ resolution — **the structure that makes the baseline strong is the structure that starves the test of signal.**

7.7 Why the field's rules hit the wall earlier: distance-testing is the wrong statistic

The algorithms of [6, 3] do not run the directional test; they threshold the plug-in distance $\hat{\ell}_1(\hat{p}_k, q)$. That class is blind far before Theorem 1(i) bites: at $k \ll r$ the empirical $\hat{p}_k$ is k atoms of mass $1/k$, so $\hat{\ell}_1 \approx 2$ *regardless of the advice's quality* (numerically: 1.908 vs 1.913 on the two canonical scenarios whose true ℓ_1 differ by 0.225). Any threshold therefore either always accepts or always rejects — the §5.3 pathology, now explained. The deeper reason is the tolerant-testing barrier: estimating $\ell_1(p, q)$ to constant accuracy at $r = \Theta(n)$

genuinely requires $\tilde{\Theta}(n/\log n)$ samples [4]. The resolution of the apparent paradox — the *decision* is easy (Theorem 1(ii)) while the *distance* is hard — is Lemma 2: the payoff is a linear, per-sample functional, and ℓ_1 is not. The practical moral for algorithm designers is one sentence: **test the payoff, not the prediction.**

We record the flip side honestly. The natural conjecture behind an earlier version of this section — that tolerant-testing hardness makes the follow/fallback decision itself impossible for *any* rule at $k = o(n/\log n)$ with $\Theta(1)$ stakes — is **refuted** by Theorem 1(ii): no witness pair with $\gamma_k = o(1)$ and $\Theta(1)$ stakes exists on this family, because by Lemma 2 a $\Theta(1)$ stake gap forces $\gamma_k \to 1$ at any $k = \omega(1)$. The tolerant-testing hard instances evade Lemma 2 only by mismatching the advice's per-cell magnitudes — precisely the regime in which distance and payoff decouple and ℓ_1 stops being the decision-relevant quantity.

7.8 Scope and honesty

Both halves of Theorem 1 use only elementary tools (Bernstein; Hellinger tensorization plus Lemma 1) — deliberately: the earlier, stronger-sounding route through tolerant-testing lower bounds is closed by §7.7, and we consider the refutation itself part of the contribution. The law is stated for decomposable (disjoint-cell) families; by Lemma 2 the payoff is per-sample observable on *every* such family, so no decomposable construction can restore a super-$1/\delta^2$ barrier. Whether a *non-decomposable* family — long-range dependence making the payoff a genuinely hard functional — can separate the budget from $1/\delta^2$ is open. The directional test is analyzed here only on the cell family; its novelty relative to payoff-estimating switching rules elsewhere in the learning-augmented literature is under a dedicated literature pass. We credit Choo et al. for the constructive baseline-coupling their threshold already exhibits; our contribution is the two-sided, rule-independent budget law, the payoff/distance separation, and the quantified wall.

8 Application Case Study: AI-Inference Serving

To show the framework instantiates a contemporary problem, we cast request routing in an AI-inference serving system as online b-matching: resources are model replicas / cache shards with a capacity c, arrivals are requests drawn from a (non-uniform, bursty) traffic distribution, an edge is a capability or cache affinity, and goodput is the competitive ratio against the b-matching optimum. We instantiate it on three real traces (Wikipedia pageviews, an Azure LLM inference trace, and the Mooncake prefix-cache trace) and across four serving concerns: capacity c, stale traffic forecasts, dynamic service times, and prefix-cache-aware routing.

We present this as a **case study, not a novelty claim** — the systems facts we recover are established, and the "with-predictions" vocabulary is a re-labeling of them. The framework does reproduce them cleanly: capacity is a *substitute* for algorithmic robustness (blind trust in a forecast degrades further as capacity grows, while a capacity-aware baseline stays safe); a live-load signal beats a stale forecast under dynamic service times; and a *stable* cache-affinity router beats a reactive one — the reverse of the traffic-forecast case. Because the literature (§9) suggested the with-predictions lens might yield a *new* actionable serving result on a tail objective, we probed it: on an SLO/tail objective under bursty load, a non-predictive policy (static headroom or reactive-adaptive reservation) matches a clairvoyant oracle to within $\leq 3\%$ across every regime we swept (`docs/SERVING_SLO_PROBE.md`). We found no regime where foresight helps — the tail objective is forgiving too, a third face of the paper's wall. Serving therefore stays a case study.

9 Related Work

Algorithm attributions are given where each algorithm is introduced (§2), and the positioning of our budget–stakes law against prior testing and tradeoff results is in §7; here we place the remaining landscape.

Learning-augmented algorithms. The consistency/robustness framework originates with competitive caching with predictions [12] and the optimal-tradeoff analyses that followed [14]; our thesis — that on average-case matching the value is robustness, not consistency — is a same-spirit but problem-specific statement, made quantitative and then proven necessary. The direct experimental-study template is Chłędowski et al. [5] in caching, whose blind-follow-with-switching combiner we benchmark (§5.4).

Online matching with predictions. MinPredictedDegree [1] and the test-and-fallback schemes [6, 3] are the algorithms we unify and, for the latter, bound; each was previously studied in isolation. Borodin et al. [2] is the advice-free experimental foundation our benchmark extends.

Distribution testing. Tolerant identity testing [4, 13] and ℓ_1-distance estimation [10] enter our story as the explanation of why the field's empirical-ℓ_1 acceptance rules are blind at sublinear prefixes — the near-linear $\tilde{\Theta}(r/\log r)$ price of tolerance is real. Our budget–stakes law shows the follow/fallback *decision* nonetheless escapes that barrier: its decision-relevant statistic is the payoff, not the distance (§7.7), which is why our lower bound is a Hellinger computation rather than a testing reduction.

Serving systems. The systems results our §8 case study recovers are established across Preble, Mooncake, SageServe, and related work; we use them as ground truth, not as contributions.

10 Limitations and Conclusion

Limitations. (i) We work in the known-i.i.d. model; since Known-IID $\leq$ Random-Order, the algorithms' guarantees carry, but our empirical wall is an average-case statement and we do not claim it for adversarial arrival order. (ii) Following the original authors, the test-and-fallback experiments use an empirical-ℓ_1 surrogate for the (unimplemented) distribution tester; the theory does *not* inherit this modeling — the law binds any rule — and §7.7 separately explains the surrogate's blindness. (iii) The degree- and histogram-prediction families do not map onto every graph, which is why we report them in parallel panels rather than one table. (iv) Each real modality is exercised by one trace. (v) The budget–stakes law is proven on decomposable (disjoint-cell) families, where the payoff identity holds; whether a *non*-decomposable family can push the budget above $1/\delta^2$ is open, as is the novelty of payoff-estimating acceptance rules relative to the broader learning-augmented literature (a dedicated pass is in progress). An earlier draft claimed a tolerant-testing impossibility for any sublinear rule; that claim was refuted during its own witness step and the refutation is reported in §7.7.

Conclusion. We gave the first unified experimental study of learning-augmented online bipartite matching — advice-free baselines, MinPredictedDegree and its augmentations, and the test-and-fallback algorithms — on one harness, with confidence intervals, across synthetic graphs, six real graphs, and real traces. Across every setting the same wall appears: the advice-free baseline is already near-optimal, unguarded prediction-following crashes below it, and the value of the sophisticated algorithms is downside protection, delivered by structural or adaptive robustness with distinct trade-offs. We then priced the wall for the test-and-fallback class with a two-sided budget–stakes law: the follow/fallback decision costs a prefix of $\tilde{\Theta}(\theta/\delta^2)$ for *any* decision rule, an explicit directional test achieves it, and — because stakes are capped by the baseline slack — on strong-baseline instances the price exceeds the horizon: upsides below $\Theta(\sqrt{(1-\rho_{\text{base}})/n})$ are uncapturable at any feasible prefix. Experiments discover the wall; theory sends the bill. The practical message is a single sentence — **on average-case matching, the advice's upside is smaller than the price of finding out whether to trust it** — with one constructive corollary for algorithm designers: *test the payoff, not the prediction* (the payoff is a per-sample observable where the distance is not). Progress on predictions for online matching will come from the regimes this paper brackets: adversarial or non-stationary arrivals, and objectives (tail, fairness, cost) where the advice-free baseline is genuinely far from optimal.

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